

SAMPLING

ENSC 429 – Digital Signal Processing

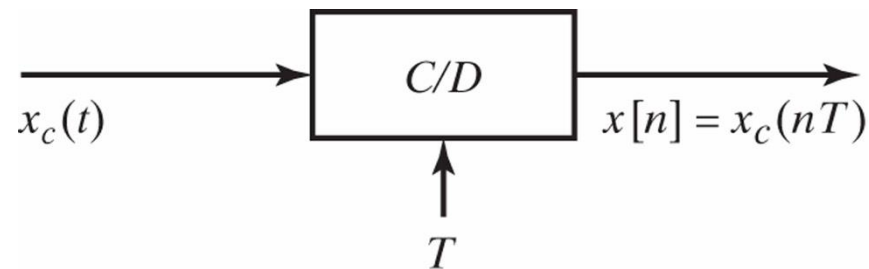
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PERIODIC SAMPLING

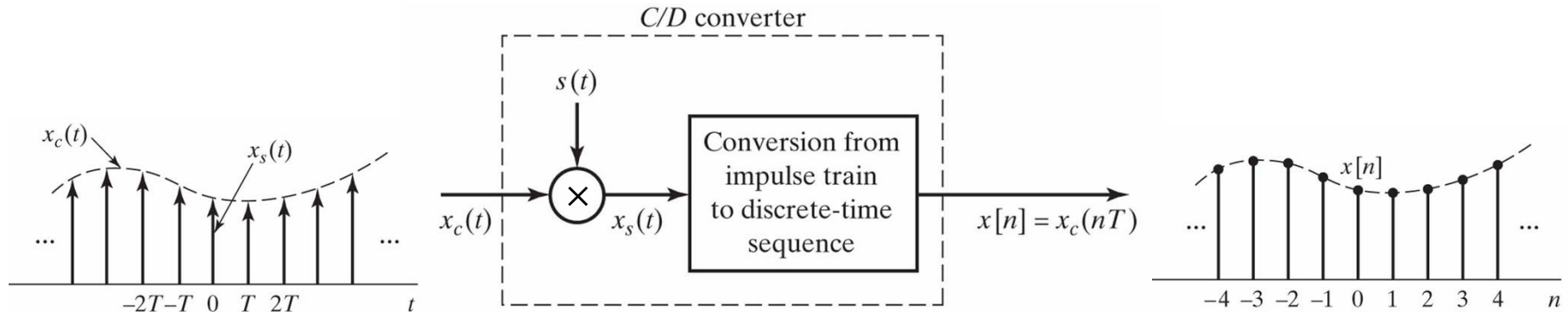
- The most common way to obtain discrete-time signals is to sample continuous-time signals at a fixed sampling rate, which is referred to as periodic sampling

$$x[n] = x_c(nT), \quad n = \dots, -2, -1, 0, 1, 2, \dots$$



- This is the ideal continuous-to-discrete-time (C/D) converter
- T is the **sampling period** in seconds
- $f_s = 1/T$ is the **sampling frequency** in samples per second
- $\Omega_s = 2\pi f_s = 2\pi/T$ is the **sampling frequency** in radians per second

MATHEMATICAL MODEL OF PERIODIC SAMPLING



$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

Periodic impulse train

$$\begin{aligned} x_s(t) &= x_c(t)s(t) = x_c(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} x_c(t)\delta(t - nT) \\ &= \sum_{n=-\infty}^{\infty} x_c(nT)\delta(t - nT) \end{aligned}$$

Non-zero only
at locations of
impulses:

$$\begin{aligned} f(t)\delta(t - t_0) \\ = f(t_0)\delta(t - t_0) \end{aligned}$$

MATHEMATICAL MODEL OF PERIODIC SAMPLING

- To understand what is happening during sampling, we have to look at the frequency domain
- $x_c(t)$ is a continuous-time signal, so its frequency-domain representation is the continuous-time Fourier transform (CTFT) $X_c(j\Omega)$

$$x_c(t) \xleftrightarrow{\text{CTFT}} X_c(j\Omega)$$

- Also, from the CTFT tables, we can find the CTFT of the impulse train:

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \xleftrightarrow{\text{CTFT}} S(j\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s)$$

- Finally, since $x_c(t)$ and $s(t)$ are multiplied in the time domain, their Fourier transforms are convolved in the frequency domain (and scaled by $1/2\pi$)

$$x_s(t) = x_c(t)s(t) \xleftrightarrow{\text{CTFT}} X_s(j\Omega) = \frac{1}{2\pi} X_c(j\Omega) * S(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$

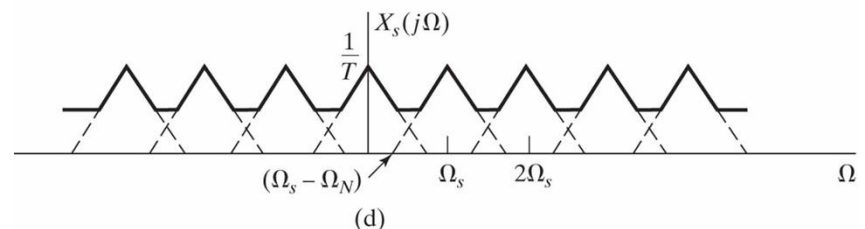
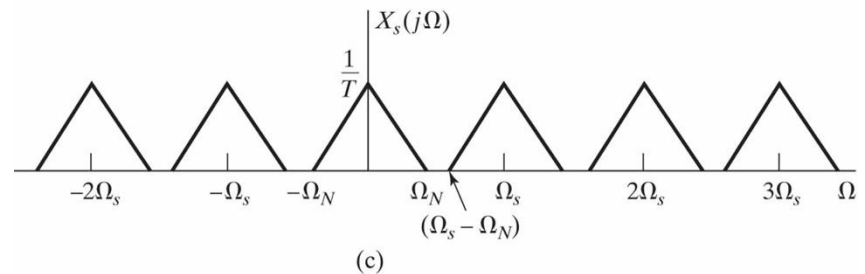
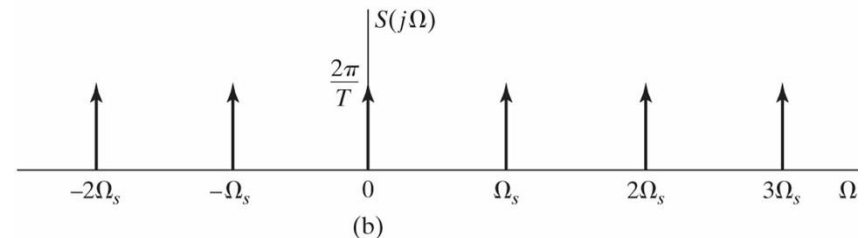
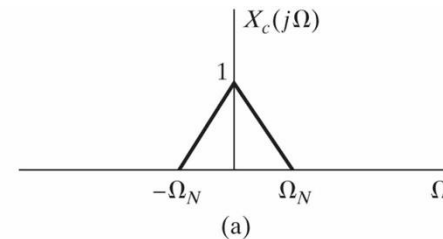
$f(t) * \delta(t - t_0) = f(t - t_0)$

MATHEMATICAL MODEL OF PERIODIC SAMPLING

- During sampling, the spectrum of the original signal, $X_c(j\Omega)$, gets replicated at the locations of impulses in the impulse train $S(j\Omega)$
- These impulses are spaced by Ω_s in the frequency domain
- Replicas of $X_c(j\Omega)$ might end up overlapping
- If Ω_N is the highest frequency in $X_c(j\Omega)$, then:

If $\Omega_N < \Omega_s - \Omega_N \Rightarrow$ no overlap

If $\Omega_N > \Omega_s - \Omega_N \Rightarrow$ overlap (aliasing)



(NYQUIST-SHANNON) SAMPLING THEOREM

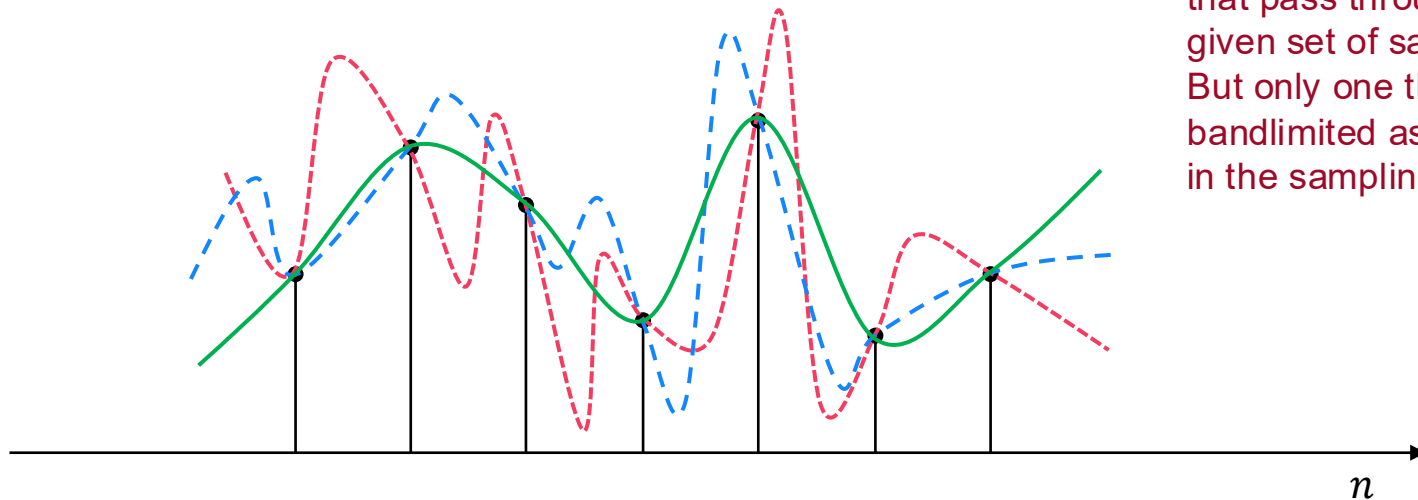
- If $x_c(t)$ is a bandlimited signal such that

$$X_c(j\Omega) = 0 \quad \text{for } |\Omega| \geq \Omega_N,$$

then $x_c(t)$ is uniquely determined by its samples $x[n] = x_c(nT)$, $n = 0, \pm 1, \pm 2, \dots$ so long as the sampling frequency satisfies

$$\Omega_s = 2\pi f_s = \frac{2\pi}{T} \geq 2\Omega_N$$

There are infinitely many continuous-time signals that pass through a given set of samples... But only one that is bandlimited as indicated in the sampling theorem.



EFFECT OF SAMPLING IN THE DTFT DOMAIN

- Consider the sampling equation

$$x_s(t) = \sum_{n=-\infty}^{\infty} x_c(nT) \delta(t - nT)$$

- Apply CTFT, remembering that $x_c(nT)$ are constants (not functions of time t), and $\delta(t - nT) \xrightarrow{\text{CTFT}} e^{-j\Omega T n}$, and $x_c(nT) = x[n]$

$$X_s(j\Omega) = \sum_{n=-\infty}^{\infty} x_c(nT) e^{-j\Omega T n} = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega T n}$$

- Compare this with the expression for DTFT of $x[n]$:

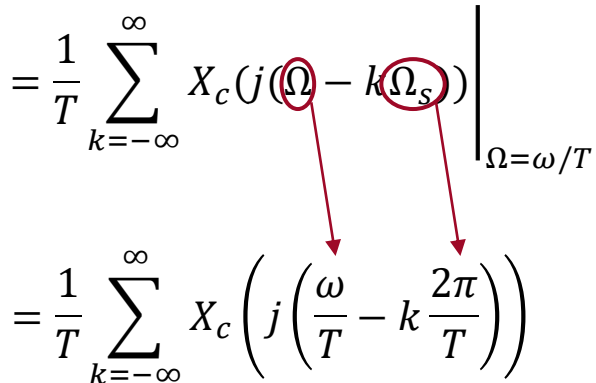
$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

- We see that

$$X_s(j\Omega) = X(e^{j\omega}) \Big|_{\omega=\Omega T}$$

EFFECT OF SAMPLING IN THE DTFT DOMAIN

- Use the derived relationship between CT frequency and DT frequency by replacing Ω by ω/T in the expression for $X_s(j\Omega)$:

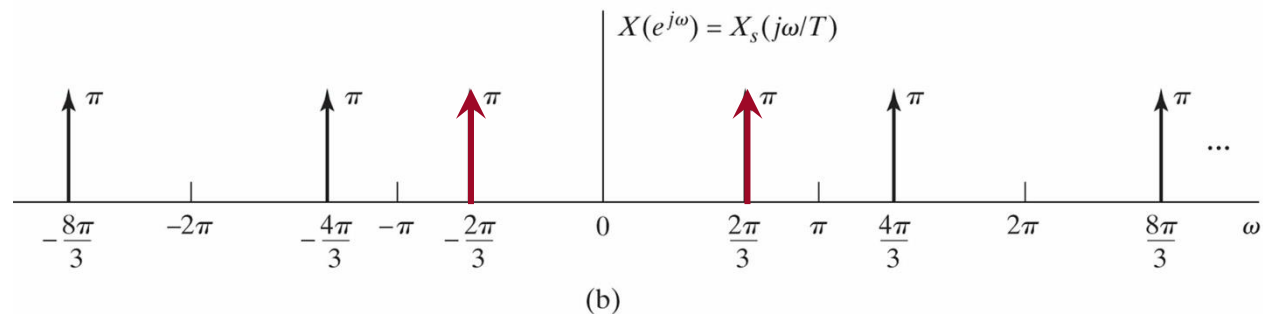
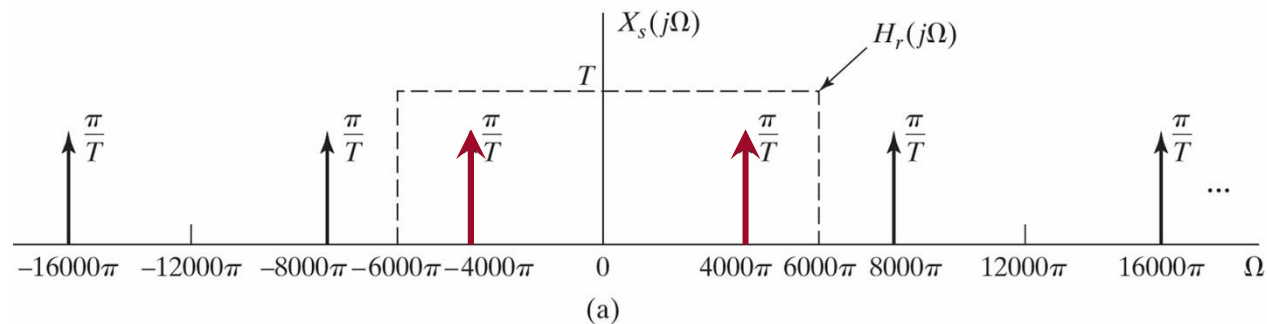
$$\begin{aligned} X(e^{j\omega}) &= X_s(j\Omega) \Big|_{\Omega=\omega/T} = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s)) \Big|_{\Omega=\omega/T} \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T} - k\frac{2\pi}{T}\right)\right) \end{aligned}$$


EXAMPLE

- Consider sampling the following sinusoidal signal with the sampling period $T = 1/6000$, then recovering it with an ideal lowpass filter with $\Omega_c = 6000\pi$:

$$x_c(t) = \cos(4000\pi t)$$

- See notes for details

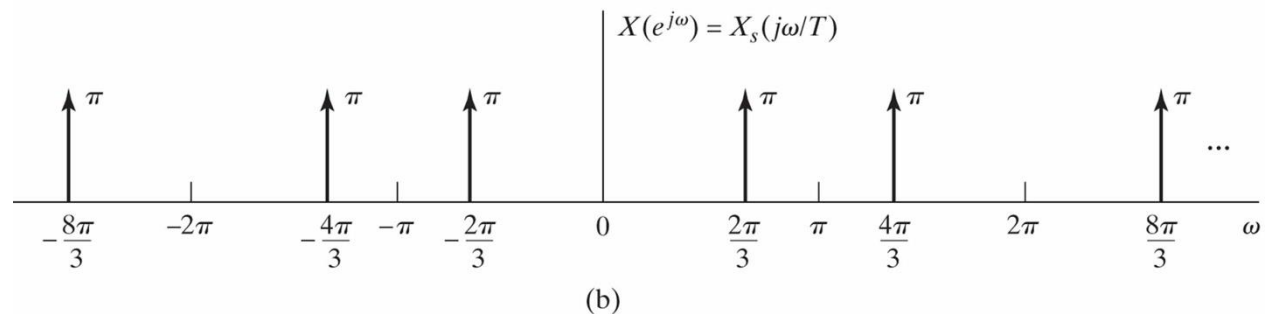
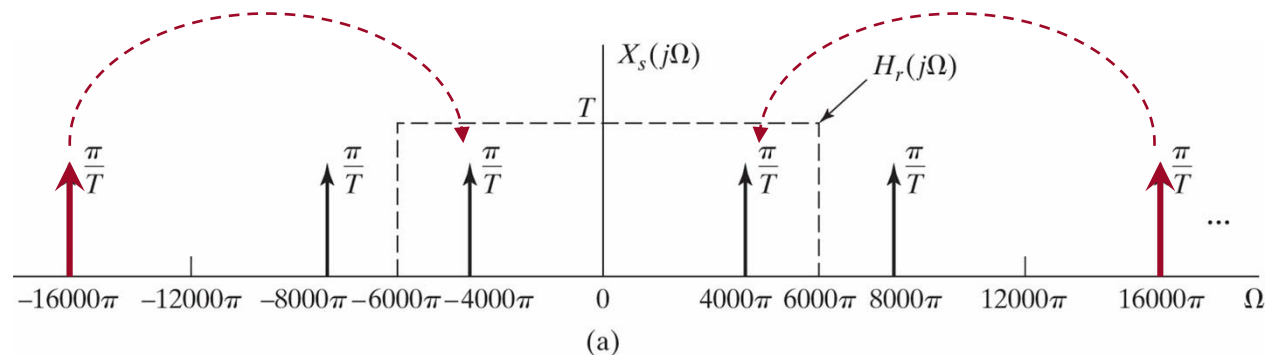


EXAMPLE

- Consider sampling the following sinusoidal signal with the sampling period $T = 1/6000$, then recovering it with an ideal lowpass filter with $\Omega_c = 6000\pi$:

$$x_c(t) = \cos(16000\pi t)$$

- See notes for details

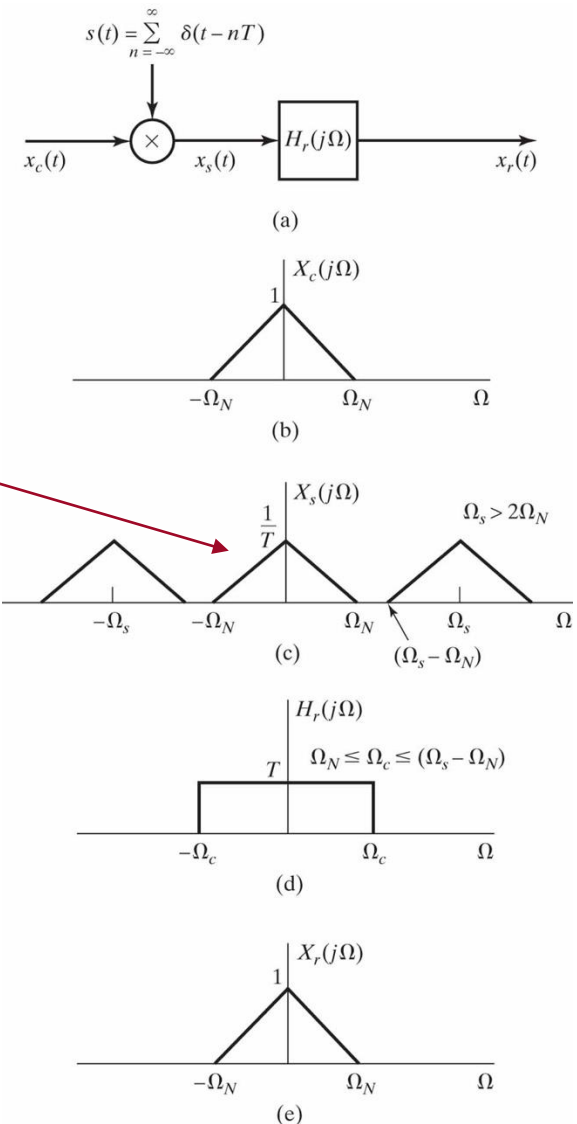


RECOVERY OF THE CT SIGNAL FROM ITS SAMPLES

- Recovery of the original continuous-time (CT) signal $x_c(t)$ from its samples can be accomplished by (ideal) lowpass filtering
- If the signal has been sampled according to the sampling theorem, the replicas of its original spectrum do not overlap; one of these replicas occupies the spectrum in the range $[-\Omega_N, \Omega_N]$
- Passing this signal through a lowpass filter whose cutoff frequency Ω_c satisfies

$$\Omega_N \leq \Omega_c \leq \Omega_s - \Omega_N$$

will recover this part of the spectrum, which matches the original spectrum



RECOVERY OF THE CT SIGNAL FROM ITS SAMPLES

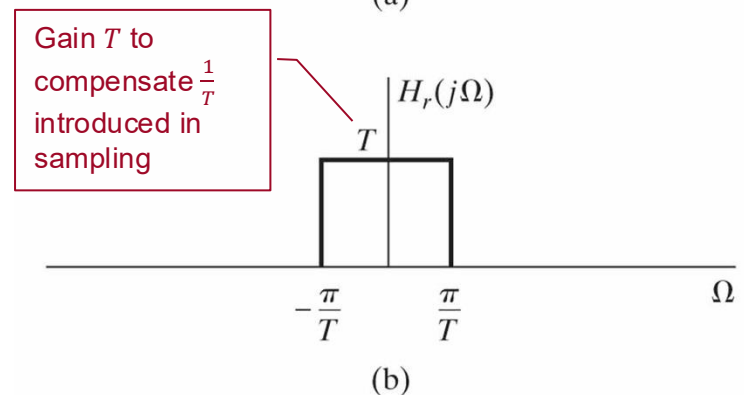
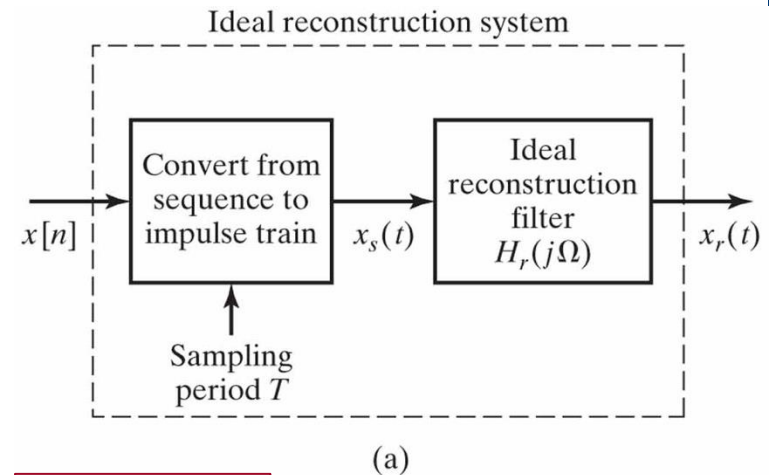
- Starting with the sequence of samples $x[n]$, first form a CT impulse train such that the elements of $x[n]$ scale the impulses

$$\begin{aligned} x_s(t) &= \sum_{n=-\infty}^{\infty} x[n] \delta(t - nT) \\ &= \sum_{n=-\infty}^{\infty} x_c(nT) \delta(t - nT) \end{aligned}$$

- Then pass through the ideal reconstruction filter with frequency response $H_r(j\Omega)$
- The cutoff frequency is usually set to half the sampling frequency:

$$\Omega_c = \frac{\Omega_s}{2} = \frac{\pi}{T}$$

- This works for any signal sampled in accordance to the sampling theorem because if $\Omega_s > 2\Omega_N$ then $\Omega_c > \Omega_N$

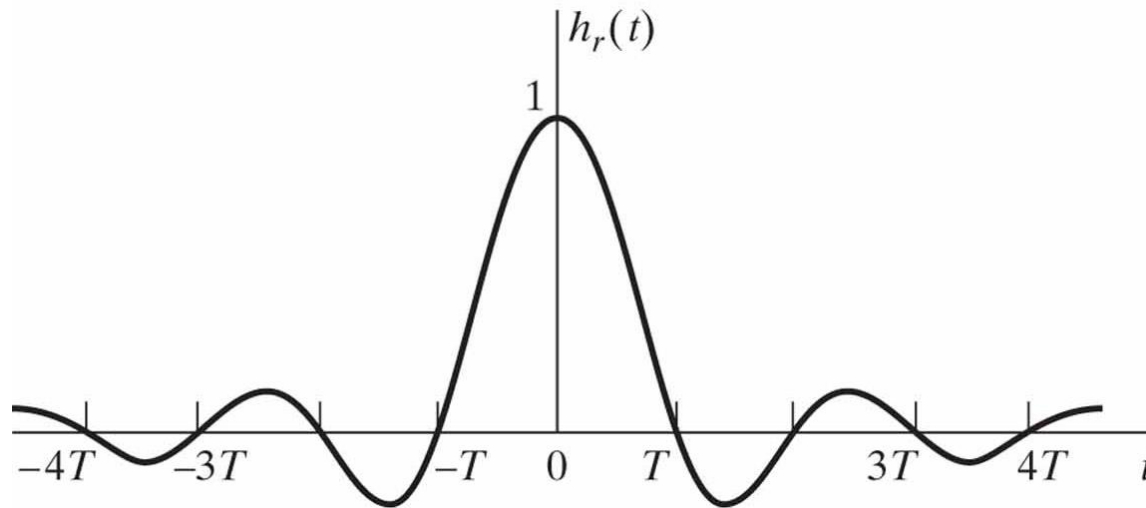


RECOVERY OF THE CT SIGNAL FROM ITS SAMPLES

- What is the impulse response of the ideal CT recovery filter?

$$h_r(t) = \text{CTFT}^{-1}\{H_r(j\Omega)\} = \frac{\sin(\pi t/T)}{\pi t/T}$$

- See notes for details



RECOVERY OF THE CT SIGNAL FROM ITS SAMPLES

- Since the recovery filter is an LTI system, when $x_s(t)$ passes through it to obtain $x_r(t)$, we have

$$x_s(t) = \sum_{n=-\infty}^{\infty} x_c(nT)\delta(t - nT) \quad \Rightarrow \quad x_r(t) = \sum_{n=-\infty}^{\infty} x_c(nT)h_r(t - nT)$$

Linear combination of delayed impulses

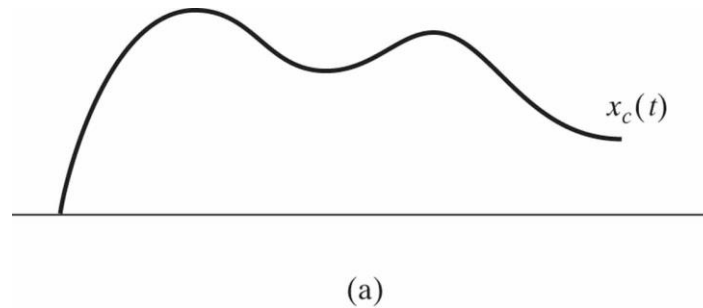
Linear combination of delayed impulse responses

- Finally

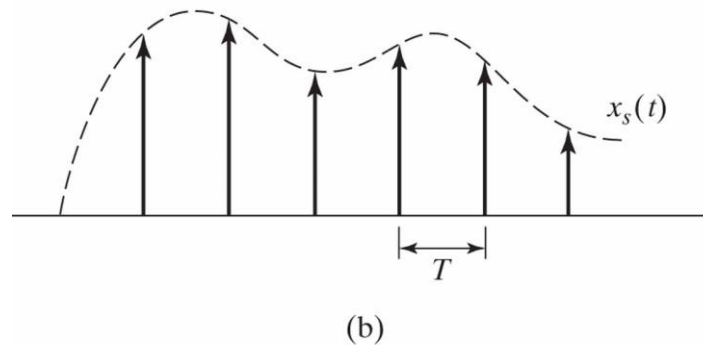
$$x_r(t) = \sum_{n=-\infty}^{\infty} x_c(nT) \frac{\sin(\pi(t - nT)/T)}{\pi(t - nT)/T}$$

RECOVERY OF THE CT SIGNAL FROM ITS SAMPLES

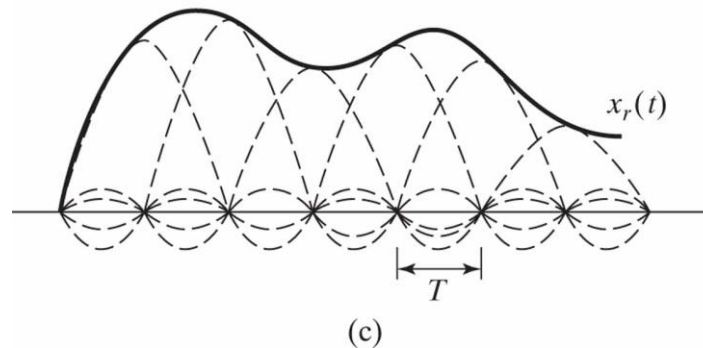
Original CT signal



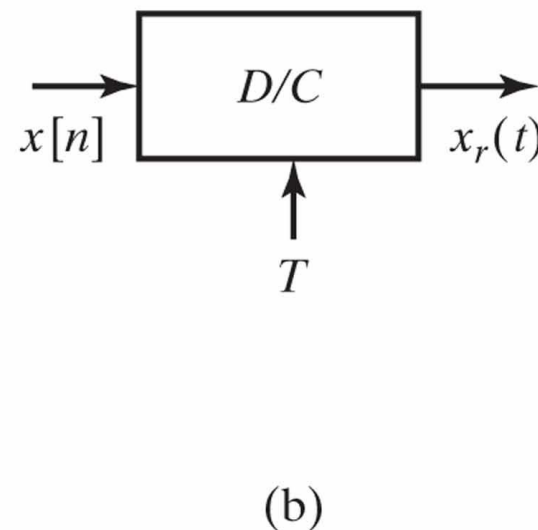
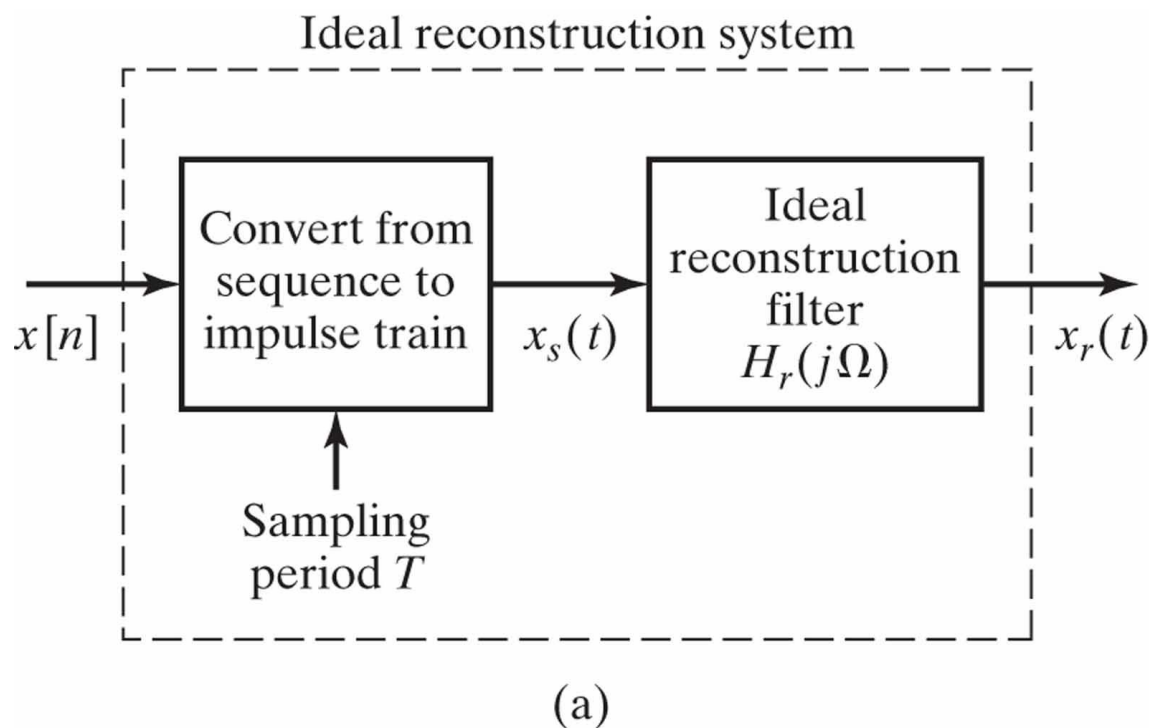
Train of scaled impulses



Recovered signal



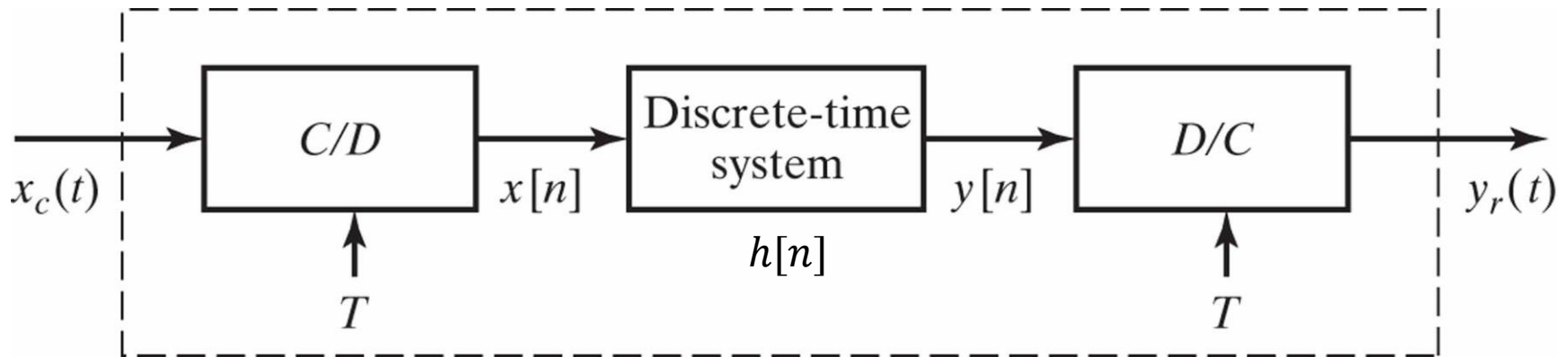
IDEAL DISCRETE-TO-CONTINUOUS TIME (D/C) CONVERTER



DT PROCESSING OF CT SIGNALS

Many applications of DSP involve DT processing of CT signals:

- Convert CT signal to DT using C/D converter
- Process the DT signal using a DT system (filter)
- Convert the result back to CT using a D/C converter



DT PROCESSING OF CT SIGNALS

- C/D converter:

$$x[n] = x_c(nT)$$

- DTFT of $x[n]$:

$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T} - k\frac{2\pi}{T}\right)\right)$$

- Output of the DT LTI system:

$$y[n] = h[n] * x[n], \quad Y(e^{j\omega}) = H(e^{j\omega}) \cdot X(e^{j\omega})$$

- Output of the D/C converter:

$$Y_r(j\Omega) = H_r(j\Omega) \cdot Y(e^{j\Omega T}) = \begin{cases} TY(e^{j\Omega T}), & |\Omega| < \pi/T \\ 0, & \text{else} \end{cases}$$

DT PROCESSING OF CT SIGNALS

- Overall, with $\omega = \Omega T$:

$$\begin{aligned} Y_r(j\Omega) &= H_r(j\Omega) \cdot Y(e^{j\Omega T}) = H_r(j\Omega) \cdot H(e^{j\Omega T}) \cdot X(e^{j\Omega T}) \\ &= H_r(j\Omega) \cdot H(e^{j\Omega T}) \cdot \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\Omega - k \frac{2\pi}{T}\right)\right) \end{aligned}$$

- Recall that

$$H_r(j\Omega) = \begin{cases} T, & |\Omega| < \pi/T \\ 0, & \text{else} \end{cases}$$

- So, if $x_c(t)$ had been sampled according to the sampling theorem and there is no aliasing, then the only frequency content within $\Omega \in \left(-\frac{\pi}{T}, \frac{\pi}{T}\right)$ is the original spectrum $X_c(j\Omega)$, i.e., the term with $k = 0$ in the above sum

$$Y_r(j\Omega) = \begin{cases} H(e^{j\Omega T}) \cdot X_c(j\Omega), & |\Omega| < \pi/T \\ 0, & \text{else} \end{cases}$$

DT PROCESSING OF CT SIGNALS

- The overall system behaves as a CT LTI system:

$$Y_r(j\Omega) = H_{\text{eff}}(j\Omega) \cdot X_c(j\Omega)$$

where $H_{\text{eff}}(j\Omega)$ is the **effective CT frequency response** of the overall system:

$$H_{\text{eff}}(j\Omega) = \begin{cases} H(e^{j\Omega T}), & |\Omega| < \pi/T \\ 0, & \text{else} \end{cases}$$

- So $H_{\text{eff}}(j\Omega)$ matches the frequency response of the DT system with $\omega = \Omega T$ over the frequency range $\Omega \in \left(-\frac{\pi}{T}, \frac{\pi}{T}\right)$, which corresponds to $\omega \in (-\pi, \pi)$
- This shows that we can design the DT system to achieve the desired CT effect

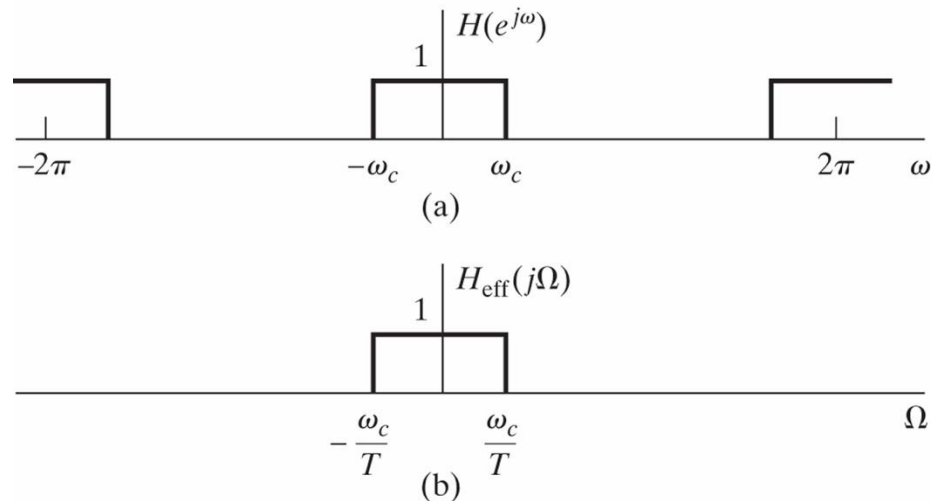
EXAMPLE

- Suppose the DT system is the ideal lowpass filter

$$H(e^{j\omega}) = \begin{cases} 1, & |\omega| < \omega_c \\ 0, & \omega_c \leq |\omega| \leq \pi \end{cases}$$

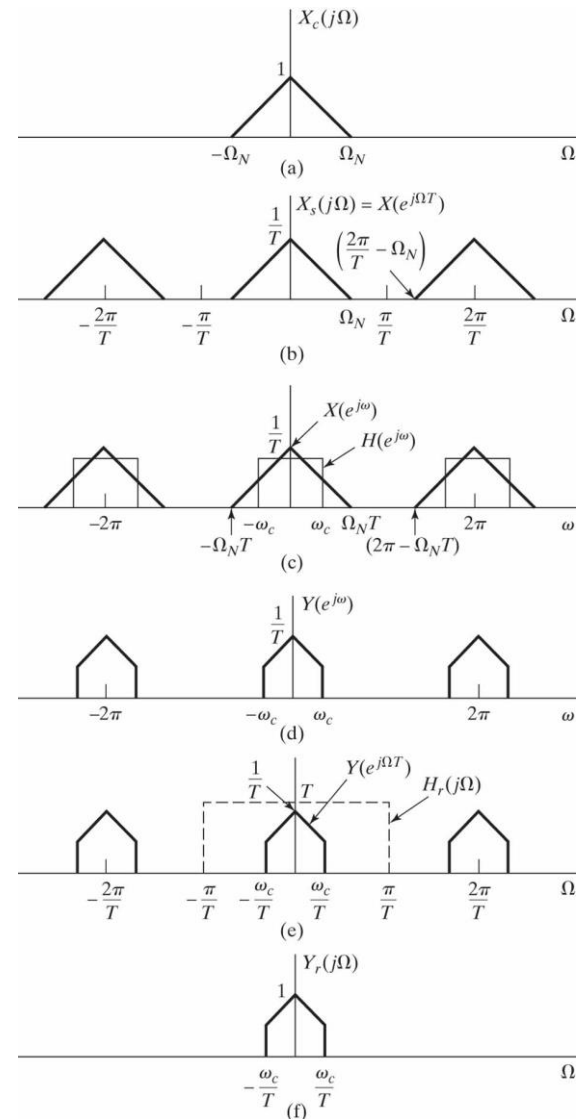
- Then the effective CT frequency response, using $\omega = \Omega T$, is

$$H_{\text{eff}}(j\Omega) = \begin{cases} 1, & |\Omega| < \omega_c/T \\ 0, & \text{else} \end{cases}$$



EXAMPLE

Frequency-domain representation at various points in the overall system



EXAMPLE

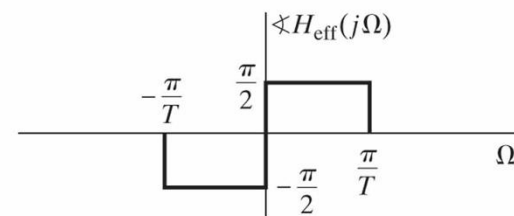
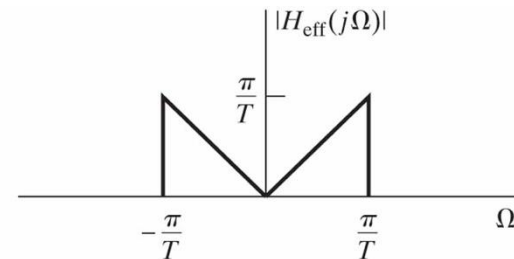
- DT implementation of an ideal CT differentiator

$$y_c(t) = \frac{d}{dt} [x_c(t)]$$

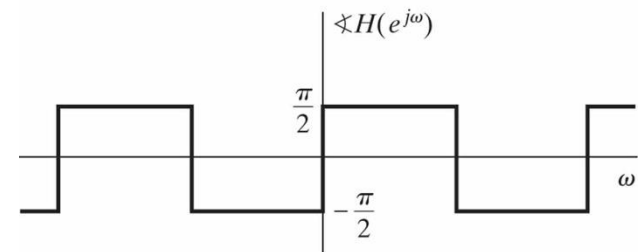
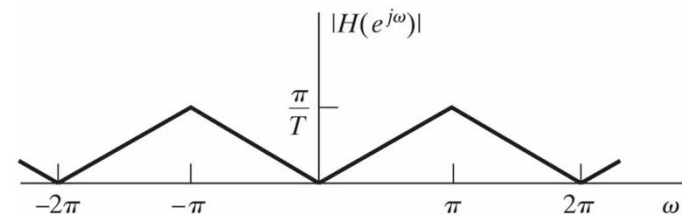
- It can be shown that the corresponding DT system has the following impulse response

$$h[n] = \begin{cases} 0, & n = 0 \\ \frac{\cos \pi n}{nT}, & \text{else} \end{cases}$$

- See notes for details



(a)



(b)

IMPULSE INVARIANCE

- Suppose we are given a CT system with impulse response $h_c(t)$ and bandlimited frequency response $H_c(j\Omega)$
- Choose the sampling interval T such that $H_c(j\Omega) = 0$ for $|\Omega| \geq \pi/T$ (i.e., in accordance with the sampling theorem)
- Let's examine what happens when we sample $h_c(t)$ as follows:

$$h[n] = T \cdot h_c(nT)$$

- From earlier analysis we know

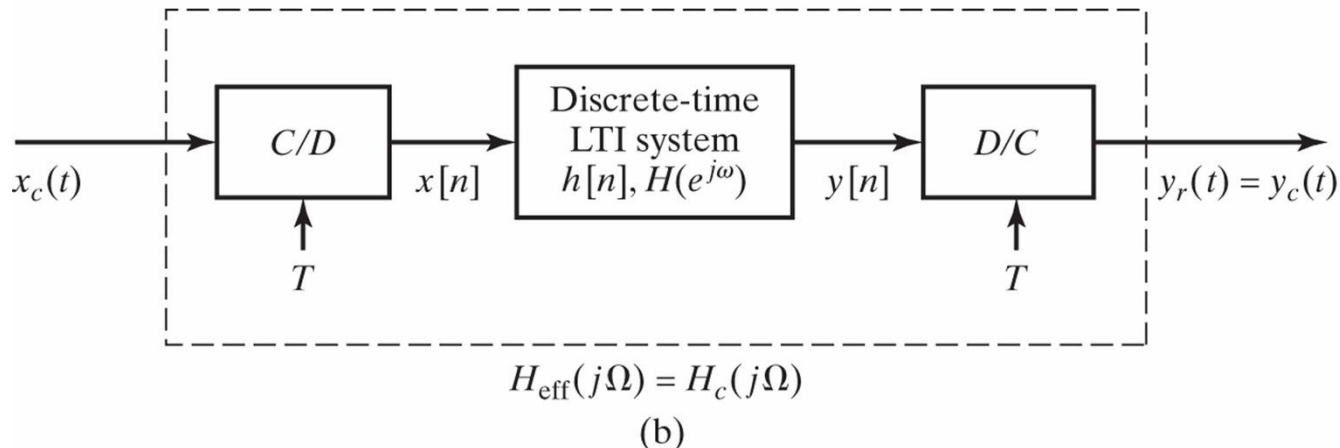
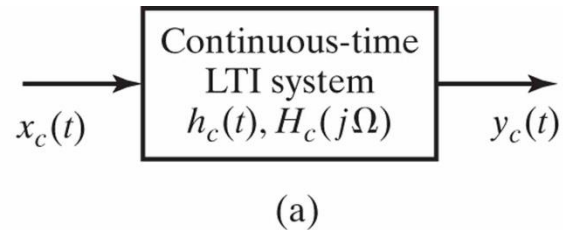
$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} H_c\left(j\left(\frac{\omega}{T} - k\frac{2\pi}{T}\right)\right)$$

- Since $H_c(j\Omega) = 0$ for $|\Omega| \geq \pi/T$, the only frequency content inside the DT frequency interval $\omega \in (-\pi, \pi)$ is due to the original spectrum (i.e., there is no aliasing), so

$$H(e^{j\omega}) = H_c\left(j\frac{\omega}{T}\right), \quad \omega \in (-\pi, \pi)$$

IMPULSE INVARIANCE

- In other words, DT filter with impulse response obtained by sampling the CT impulse response $h_c(t)$ will have the same frequency response as the CT filter
- Also, this is one of the ways to design DT filters



EXAMPLE

- Example: design an ideal DT lowpass filter with cutoff frequency $\omega_c < \pi$
- We can start with an ideal CT lowpass filter

$$H_c(j\Omega) = \begin{cases} 1, & |\Omega| < \Omega_c \\ 0, & \text{else} \end{cases}$$

- Then sample its impulse response
- [See notes](#) for details

READING ASSIGNMENT

- Read Sections 4.1 – 4.4 in the textbook
- Start reading Chapter 5